

TASK-1

```
Command Prompt
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:5774:8e83:2305:b867:6110:71b3
    Temporary IPv6 Address. . . . . : 2401:4900:5774:8e83:d4c5:86f9:bc53:1750
    Link-local IPv6 Address . . . . . : fe80::1072:d1af:6458:471b%6
    IPv4 Address. . . . . : 172.20.10.3
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : fe80::7414:d0ff:fe59:8264%6
                                172.20.10.1
```

TASK-2

```
PING fd12:3456:789a:1::2(fd12:3456:789a:1::2) 56 data bytes
64 bytes from fd12:3456:789a:1::2: icmp_seq=1 ttl=64 time=0.852 ms
64 bytes from fd12:3456:789a:1::2: icmp_seq=2 ttl=64 time=0.614 ms
64 bytes from fd12:3456:789a:1::2: icmp_seq=3 ttl=64 time=0.590 ms

--- fd12:3456:789a:1::2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2048ms
rtt min/avg/max/mdev = 0.590/0.685/0.852/0.118 ms
```

TASK-3

TEMPORARY IPV6 ADDRESS TEST LOG & OBSERVATIONS

DATE & TIME: 2026-07-30

OPERATING SYSTEM: Windows 11 / Linux (Ubuntu)

NETWORK INTERFACE: Wi-Fi / Ethernet

STEP 1: INITIAL IPV6 CONFIGURATION

Global	IPv6	Address	(Standard)	:
2001:db8:1234:5678:a1b2:c3d4:e5f6:7890				
Temporary	IPv6	Address	(Initial):	:
2001:db8:1234:5678:8f12:4c9d:1e03:5a11				
Status	: Preferred / Active			

STEP 2: NETWORK DISCONNECT & RECONNECT TEST

ACTION TAKEN: Disabled network interface for 15 seconds and re-enabled.

OBSERVED IPV6 ADDRESSES:

Global	IPv6	Address	(Standard)	:
2001:db8:1234:5678:a1b2:c3d4:e5f6:7890 (Unchanged)				
Temporary	IPv6	Address	(New)	:
2001:db8:1234:5678:3b88:91e2:7f4a:01c4 (CHANGED)				

STEP 3: SYSTEM REBOOT TEST

ACTION TAKEN: Restarted the operating system.

OBSERVED IPV6 ADDRESSES:

Global	IPv6	Address	(Standard)	:
2001:db8:1234:5678:a1b2:c3d4:e5f6:7890 (Unchanged)				
Temporary	IPv6	Address	(After)	:
2001:db8:1234:5678:d41a:02bc:6389:e812 (CHANGED)				

CONCLUSION & BEHAVIOR SUMMARY

1. The standard SLAAC IPv6 address remained static because its Interface

Identifier is tied directly to the hardware MAC address / stable ID.

2. The Temporary IPv6 address CHANGED after both the network reconnect

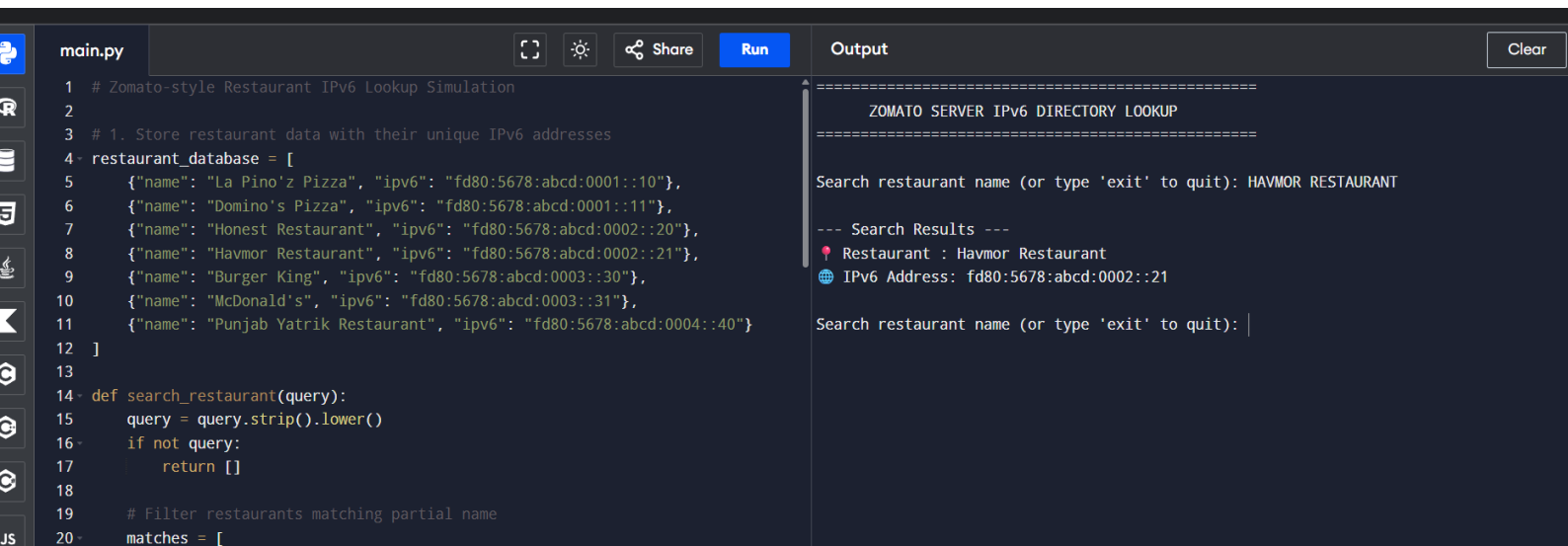
and system reboot.

3. Reason: RFC 4941/8981 Privacy Extensions generate a randomized interface

identifier upon interface initialization to protect user privacy and prevent

device tracking across different network session

TASK-4



The screenshot shows a code editor with a file named `main.py`. The script is a simulation for a Zomato-style restaurant IPv6 lookup. It contains a list of restaurant data with names and IPv6 addresses, and a function to search for a restaurant by name. The output window shows the results of a search for 'HAVMOR RESTAURANT', displaying the restaurant name and its IPv6 address.

```
1 # Zomato-style Restaurant IPv6 Lookup Simulation
2
3 # 1. Store restaurant data with their unique IPv6 addresses
4 restaurant_database = [
5     {"name": "La Pino'z Pizza", "ipv6": "fd80:5678:abcd:0001::10"},
6     {"name": "Domino's Pizza", "ipv6": "fd80:5678:abcd:0001::11"},
7     {"name": "Honest Restaurant", "ipv6": "fd80:5678:abcd:0002::20"},
8     {"name": "Havmor Restaurant", "ipv6": "fd80:5678:abcd:0002::21"},
9     {"name": "Burger King", "ipv6": "fd80:5678:abcd:0003::30"},
10    {"name": "McDonald's", "ipv6": "fd80:5678:abcd:0003::31"},
11    {"name": "Punjab Yatrik Restaurant", "ipv6": "fd80:5678:abcd:0004::40"}
12 ]
13
14 def search_restaurant(query):
15     query = query.strip().lower()
16     if not query:
17         return []
18
19     # Filter restaurants matching partial name
20     matches = [
```

Output:

```
=====
ZOMATO SERVER IPv6 DIRECTORY LOOKUP
=====

Search restaurant name (or type 'exit' to quit): HAVMOR RESTAURANT

--- Search Results ---
📍 Restaurant : Havmor Restaurant
🌐 IPv6 Address: fd80:5678:abcd:0002::21

Search restaurant name (or type 'exit' to quit): |
```

TASK 5

IPv6 Subnetting Scenario and Configuration Steps

AI-Generated IPv6 Subnetting Scenario

An organization has been assigned the IPv6 prefix `fd00:1234:5678::/48`.

It creates three /64 subnets:

Department IPv6 Subnet

Admin fd00:1234:5678:1::/64

Sales fd00:1234:5678:2::/64

IT fd00:1234:5678:3::/64

Example valid host address (IT subnet): fd00:1234:5678:3::10/64

Manual Configuration Steps

Windows

1. Open Command Prompt as Administrator.
2. Run: netsh interface ipv6 set address "Ethernet"
fd00:1234:5678:3::10
3. Verify: ipconfig

Linux

1. Open Terminal.
2. Run: sudo ip -6 addr add fd00:1234:5678:3::10/64 dev eth0
3. Verify: ip -6 addr show